

1. Administrative & Study Identification

**Full Study Title:** An Open-label Phase 1 IV Study to Evaluate Metabolism, Excretion, and Mass Balance of [14C]MK-5475 in Healthy Participants **Short Title/Acronym:** Frespaciguat (MK-5475) Mass Balance Study **Protocol Number:** 5475-011 (CTR20212315) **NCT Number:** NCT06777602 **Sponsor Name:** Merck Sharp & Dohme Corp. **Investigational Product:** [<sup>14</sup>C]frespaciguat (MK-5475), an inhaled soluble guanylate cyclase (sGC) stimulator **Phase of Development:** Phase 1 **Medical Monitor:** Ednan K. Bajwa, MD, MPH (Translational Medicine, Merck Research Laboratories)

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To characterize the pharmacokinetics (PK), metabolism, elimination, safety, and tolerability of frespaciguat in healthy male participants following a single intravenous dose of [<sup>14</sup>C]frespaciguat. **Secondary Objectives:** To perform a human mass balance assessment to determine the predominant routes of excretion (e.g., biliary-fecal route) and the role of metabolism (e.g., via  $\beta$ -oxidation enzymes) in the clearance of the drug.

**2.2 Background & Rationale** Frespaciguat (MK-5475) is an inhaled soluble guanylate cyclase stimulator being developed to treat pulmonary hypertension associated with chronic obstructive pulmonary disease (PH-COPD). Current treatments for PH-COPD are limited, and systemic vasodilators often cause adverse effects. Frespaciguat is designed to selectively reduce pulmonary vascular pressures with minimal systemic exposure. Evaluating its excretion balance, pharmacokinetics, and metabolism through an intravenous microtracer study is a critical step in clinical development to understand its safety profile and potential for drug-drug interactions.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Single-arm (Non-comparative) **Blinding:** Open-label **Randomization:** N/A

**3.2 Planned Enrollment & Duration Target N\$:** Small cohort of healthy male participants **Number of Sites:** Single Phase 1 Clinical Pharmacology Unit **Estimated Study Start:** Completed **Estimated Primary Completion:** Completed

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Healthy male participants. 2. Adult age. 3. Capable of providing informed consent and adhering to in-clinic confinement for mass balance sample collection.

**4.2 Exclusion Criteria** 1. Female participants. 2. Any unstable medical condition or clinically significant cardiovascular disease. 3. Abnormal clinical laboratory values that would impact safety or PK analysis.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Single 100 µg dose of [<sup>14</sup>C]frespaciguat using a low radioactive dose. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Single administration followed by a multi-day confinement for specimen collection.

**5.2 Concomitant & Prohibited Therapy Permitted:** None generally permitted in healthy participant mass balance studies to avoid PK interference. **Prohibited:** Inhibitors or inducers of β-oxidation enzymes (due to their role in frespaciguat's extensive metabolism).

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## 6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures                                                                 |
|----------|---------------|-------------------------------------------------------------------------------------------|
| 1        | Day -28 to -1 | Informed Consent, Medical History, Baseline Labs                                          |
| 2        | Day 1         | Single IV dose of [ <sup>14</sup> C]frespaciguat, intensive PK blood sampling             |
| 3        | Days 2-X      | Continued PK sampling, continuous collection of urine and feces for mass balance recovery |
| 4        | Variable      | Final safety evaluation and discharge once target radioactivity recovery is achieved      |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Pharmacokinetics (PK) parameters including clearance and half-life (noted as approx. 2 hours), and overall safety/tolerability assessed via adverse events. **Measurement Method:** Serial PK blood sampling, vital signs, ECG, and standard clinical laboratory analysis.

**7.2 Secondary & Exploratory Endpoints Mass Balance/Excretion:** Proportion of radiolabeled drug eliminated via the biliary-fecal route vs. renal clearance. **Metabolism:** Identification of major metabolites (primarily driven by β-oxidation).

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Continuous monitoring during the in-clinic confinement period; standard collection of clinical laboratory tests. **Serious Adverse Events (SAEs):** Reported immediately as per regulatory requirements. No severe adverse events were reported in this study. **Data Monitoring Committee (DMC):** Internal safety review by the Sponsor (Merck).

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Pharmacokinetic (PK) Set (all subjects with evaluable PK data); Safety Set (all subjects who received the dose). **Sample Size Justification:** Small sample size appropriately powered for a descriptive mass balance evaluation without formal hypothesis testing. **Handling of Missing Data:** Standard PK non-compartmental analysis (NCA) imputation rules for concentration values below the limit of quantification.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous onsite monitoring and sample tracking during the confinement period at the Phase 1 unit. **Audit Trail:** eCRF locking, tracking of radiolabeled sample integrity, and standard data clarification protocols.

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## 11. Study Identifiers & Governance

**Primary ID:** 5475-011 **Registry Number:** NCT06777602 **Sponsor/Lead Organization:** Merck Sharp & Dohme Corp. **Study Title:** An Open-label Phase 1 IV Study to Evaluate Metabolism, Excretion, and Mass Balance of [14C]MK-5475 in Healthy Participants **Official Regulatory Title:** An Open-label Phase 1 IV Study to Evaluate Metabolism, Excretion, and Mass Balance of [14C]MK-5475 in Healthy Participants

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## 12. Clinical Framework (Study Specific)

**Primary Condition:** Healthy Participants (Target drug indication: PH-COPD) **Diagnosis Threshold:** N/A (Healthy volunteers) **Medical Methodology:** Phase 1 Mass Balance and Pharmacokinetics **Optimization Target:** Full recovery of radioactive dose to map metabolic pathways

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### 13. Study Procedures & Methodology

**Management Pathway:** In-clinic confinement for PK/ADME sampling  
**Education Protocol (HCP):** Site initiation and protocol training on handling radioactive specimens  
**Education Protocol (Patient):** Confinement rules and sample collection instructions  
**Quality Audit Mechanism:** Clinical Research Organization (CRO) monitoring and radioactive mass accountability

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### 14. Observation & Follow-Up Schedule

**Total Duration:** Until complete recovery of radioactivity (typically 1-2 weeks)  
**Onsite Visit Cadence:** Continuous confinement post-dose for specimen collection  
**Remote Monitoring Frequency:** N/A  
**Checklist Review Interval:** Daily clinical assessments and AE monitoring during confinement

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### 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|-----|-----|------|-----|
| Role                     | Name   | Signature | Date          | :-- | :-- | :--  | :-- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |     |     |      |     |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |     |     | Lead |     |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |     |     |      |     |